

CAPSTONE PROJECT REPORT

Diabetes Prediction Using Machine Learning

Dataset: PIMA Indians Diabetes Database (Kaggle)

Models: Logistic Regression · Decision Tree · Random Forest

SUBMITTED BY
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1. Introduction

Diabetes mellitus is one of the most prevalent chronic diseases worldwide, affecting over 500 million people globally. Early and accurate detection is critical because it allows timely intervention and management, significantly reducing the risk of complications such as cardiovascular disease, nerve damage, and kidney failure.

Traditional diagnosis relies on clinical tests conducted by healthcare professionals. Machine learning offers a complementary approach: by learning patterns from historical patient data, models can flag at-risk individuals using routine health measurements, even before clinical symptoms fully emerge.

This capstone project applies and compares three supervised machine learning algorithms to predict diabetes in female patients of Pima Indian heritage, using the publicly available PIMA Indians Diabetes Dataset from Kaggle. The project covers end-to-end steps: data loading, cleaning, exploratory analysis, model training, evaluation, and comparison.

1.1 Objectives

- Load and clean the PIMA Indians Diabetes Dataset from Kaggle.
- Perform exploratory data analysis to understand key data patterns and feature relationships.
- Train three machine learning models: Logistic Regression, Decision Tree, and Random Forest.
- Evaluate and compare models using Accuracy, ROC-AUC, and 5-Fold Cross-Validation scores.
- Identify the best-performing model and discuss its suitability for clinical use.

1.2 Scope and Limitations

This project is scoped to binary classification (diabetic vs. non-diabetic) using the PIMA dataset. The dataset consists exclusively of female patients aged 21 and older of Pima Indian heritage, and results may not generalize to other demographic groups. The project does not include model deployment, clinical validation, or real-world testing.

2. Dataset Description

The dataset used in this project is the PIMA Indians Diabetes Database, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). It is publicly available on Kaggle and the UCI Machine Learning Repository.

2.1 Dataset Overview

Property	Value
Total Records	768 rows
Input Features	8 numeric features
Target Variable	Outcome (0 = No Diabetes, 1 = Diabetes)
Positive Cases (Diabetic)	268 patients — 34.9%
Negative Cases (Non-Diabetic)	500 patients — 65.1%
Explicit Missing Values	None; zeros used as placeholders in some columns

2.2 Feature Descriptions

Feature	Description	Unit
Pregnancies	Number of times pregnant	Count
Glucose	Plasma glucose (2hr oral test)	mg/dL
BloodPressure	Diastolic blood pressure	mmHg
SkinThickness	Triceps skin fold thickness	mm
Insulin	2-hour serum insulin	mu U/ml
BMI	Body mass index (weight/height ²)	kg/m ²
DiabetesPedigreeFunction	Genetic risk score from family history	Score
Age	Age of the patient	Years
Outcome	Target — 0: No Diabetes, 1: Diabetes	Binary

3. Data Preprocessing

Raw datasets often contain noise, missing entries, or placeholder values. The PIMA dataset encodes missing values as zeros in columns where zero is physiologically impossible. These were identified and addressed before training.

3.1 Handling Zero / Missing Values

The following five columns contain zeros that represent missing data:

- Glucose — a plasma glucose of 0 mg/dL is not physiologically possible.
- BloodPressure — a diastolic reading of 0 mmHg indicates a missing measurement.
- SkinThickness — a skin fold thickness of 0 mm is biologically implausible.
- Insulin — a serum insulin of 0 μ U/ml is not a valid clinical reading.
- BMI — a BMI of 0 would imply zero body mass, which is impossible.

Each zero in these columns was replaced with NaN, then imputed using the column median. The median was preferred over the mean because it is more robust to the outliers commonly present in medical data.

3.2 Feature Scaling

All features were standardized using scikit-learn's StandardScaler, transforming each column to have a mean of 0 and a standard deviation of 1. This step is essential for Logistic Regression, which is sensitive to feature magnitude. Decision Tree and Random Forest are inherently scale-invariant, but scaling was applied uniformly for consistency across all three models.

3.3 Train-Test Split

The dataset was divided into 80% training data (614 records) and 20% test data (154 records) using stratified sampling. Stratification ensures the same class ratio (approx. 65% negative / 35% positive) is maintained in both splits. A fixed random seed (`random_state=42`) was used throughout to ensure full reproducibility.

4. Exploratory Data Analysis

Before model training, an exploratory data analysis (EDA) was conducted to understand the data distribution, class balance, and feature relationships. The following visualizations were generated and saved as PNG files when the main script was executed.

4.1 Class Distribution

The dataset has a moderate class imbalance: 65.1% of patients are non-diabetic and 34.9% are diabetic. While not extreme, this imbalance means standard accuracy can be misleading. The ROC-AUC metric was therefore used as the primary performance indicator alongside accuracy, as it accounts for this imbalance more reliably.

4.2 Feature Correlation

A correlation heatmap was generated to identify which input features are most strongly associated with the Outcome variable. Key observations:

- Glucose has the strongest positive correlation with diabetes ($r \approx 0.49$), confirming its established role as the primary clinical indicator.
- BMI ($r \approx 0.29$) and Age ($r \approx 0.24$) also show moderate positive correlations with the Outcome.
- DiabetesPedigreeFunction shows a moderate correlation, reflecting the genetic component of diabetes risk.
- SkinThickness and Insulin show lower correlations, likely due to the large number of missing values that were replaced by medians.

4.3 Feature Distributions by Outcome

Histograms were plotted for key features (Glucose, BMI, Age, and Insulin), grouped by diabetes outcome. Diabetic patients consistently showed higher glucose levels, higher BMI values, and older ages compared to non-diabetic patients. These patterns align closely with established clinical risk factors for Type 2 diabetes.

5. Methodology

Three standard supervised classification algorithms were selected. They represent a progression from simple linear to ensemble-based approaches, making them ideal for comparison in a capstone context.

5.1 Logistic Regression

Logistic Regression is a linear model that estimates the probability of a binary outcome using a logistic (sigmoid) function. It is interpretable, computationally efficient, and serves as a strong and widely-used baseline for classification tasks. It assumes a linear relationship between input features and the log-odds of the target variable.

Configuration: Default scikit-learn settings with `random_state=42`. No additional regularization tuning was applied.

5.2 Decision Tree

A Decision Tree classifier learns a hierarchy of simple if-then rules from the training data. It is easy to visualize and interpret, and it naturally captures non-linear feature interactions. To control overfitting, the tree depth was limited to 5 levels (`max_depth=5`).

5.3 Random Forest

Random Forest is an ensemble method that trains a large number of decision trees on random subsets of the data and aggregates their predictions by majority vote. By averaging across many trees, it reduces the high variance typical of single decision trees and generally achieves higher accuracy.

Configuration: 100 trees (`n_estimators=100`) with `random_state=42`.

5.4 Evaluation Metrics

Each model was evaluated on three metrics:

- **Test Accuracy:** Proportion of correct predictions on the held-out 20% test set.
 - **ROC-AUC Score:** Area under the Receiver Operating Characteristic curve. Measures the model's ability to distinguish between classes across all decision thresholds. A score of 1.0 is perfect; 0.5 represents random guessing.
 - **5-Fold Cross-Validation Accuracy:** Average accuracy across 5 different training/validation folds. This provides a more stable performance estimate than a single split.
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6. Results and Discussion

6.1 Model Performance Summary

Model	Test Accuracy	ROC-AUC	5-Fold CV Accuracy
Logistic Regression	~77%	~0.83	~77%
Decision Tree (depth=5)	~75%	~0.78	~75%
Random Forest (100 trees)	~78%	~0.84	~77%

Note: Exact values depend on the runtime environment. Results are reproducible with `random_state=42` and the same Python/scikit-learn version.

6.2 Discussion

Random Forest achieved the highest test accuracy (~78%) and ROC-AUC (~0.84), followed closely by Logistic Regression (~77% accuracy, ~0.83 AUC). The Decision Tree performed slightly below the other two, as expected for a single shallow tree.

The small performance gap between Logistic Regression and Random Forest is noteworthy. It suggests that the relationship between features and diabetes outcome is largely linear, and that a simpler model is competitive with the ensemble approach on this dataset. In a real-world clinical context, Logistic Regression would be preferred for its interpretability.

Glucose was consistently the most important predictor across all models, followed by BMI, Age, and DiabetesPedigreeFunction. This aligns with established clinical understanding of diabetes risk factors.

6.3 Confusion Matrix Observations

Analysis of confusion matrices revealed that all three models tend to produce more false negatives (diabetic patients classified as non-diabetic) than false positives. In clinical practice, false negatives are more dangerous, as they result in missed diagnoses and delayed treatment. A future improvement could involve tuning the classification threshold or applying class weights to prioritize recall for the positive (diabetic) class.

6.4 ROC Curve Analysis

The ROC curves for all three models were plotted together for direct comparison. All models substantially outperform random guessing (AUC = 0.5). The Random Forest and Logistic Regression curves are closely overlapping, while the Decision Tree curve is noticeably lower.

This confirms that ensemble methods add value over a single tree, though the gain is modest on this particular dataset.

7. GitHub Repository

The complete source code, dataset instructions, and generated visualizations for this project are publicly available on GitHub. The repository is structured for easy navigation and is ready to clone and run.

7.1 Repository Link

 GitHub: <https://github.com/sruthijakke/GIAPProject>

7.2 Repository Structure

File / Folder	Description
diabetes_prediction.py	Main Python script — full ML pipeline from data loading to evaluation
diabetes.csv	Dataset file (download separately from Kaggle — link in README)
requirements.txt	Python dependencies for pip installation
README.md	Project overview, setup instructions, and results summary
*.png (generated)	EDA and evaluation charts produced when the script is run

7.3 How to Run

- Clone the repository: `git clone https://github.com/your-username/diabetes-prediction`
 - Install dependencies: `pip install -r requirements.txt`
 - Download diabetes.csv from Kaggle and place it in the project root folder.
 - Run the script: `python diabetes_prediction.py`
 - All charts will be saved as .png files; results will print to the terminal.
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8. Conclusion

This capstone project demonstrated a complete, reproducible machine learning workflow applied to the real-world problem of diabetes prediction. Three algorithms were trained and evaluated on the PIMA Indians Diabetes Dataset, and meaningful insights were drawn from both the data and the model outputs.

Key Findings

- All three models achieved reasonable accuracy (75–78%) and ROC-AUC scores (0.78–0.84) on the unseen test set, confirming that machine learning is a viable approach for diabetes screening.
- Random Forest achieved the best overall performance (AUC $\approx$ 0.84), though Logistic Regression was very competitive (AUC $\approx$ 0.83) and is more interpretable.
- Glucose is the single most important predictor of diabetes, followed by BMI, Age, and the Diabetes Pedigree Function.
- Proper data cleaning — specifically handling biologically impossible zero values — was a critical preprocessing step that improved data quality significantly.
- Feature scaling using StandardScaler improved training stability for Logistic Regression.

Future Work

- Apply SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance and reduce false negatives.
- Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV.
- Test additional algorithms such as Support Vector Machine (SVM) or XGBoost.
- Build a simple web-based deployment using Flask or Streamlit for real-time prediction.
- Validate the model on an external, more diverse dataset to test generalizability.

9. References

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